

Solubility

Soluble Salt	Insoluble Salt
All Group 1 Salts	
All NH_4^+ Salts	
All NO_3^- Salts	
Most Halide	Haldie salt of Pb^{2+} , Ag^+ , Cu^+
Most Sulphate	BaSO_4 (test for SO_4^{2-}) PbSO_4 SrSO_4 Ag_2SO_4 CaSO_4 (sparingly soluble)
Group 1 & NH_4 Carbonates (mentioned above)	Most Carbonates
Group 1 & NH_4 Hydroxide $\text{Ba}(\text{OH})_2$ $\text{Sr}(\text{OH})_2$ Hydroxide	Most Hydroxides

Group II Solubility:

Sulphate **decrease** solubility down the group (BaSO_4 (s))

Group II Carbonate **all** insoluble

Hydroxide **increase** solubility down the group ($\text{Mg}(\text{OH})_2$ (s))